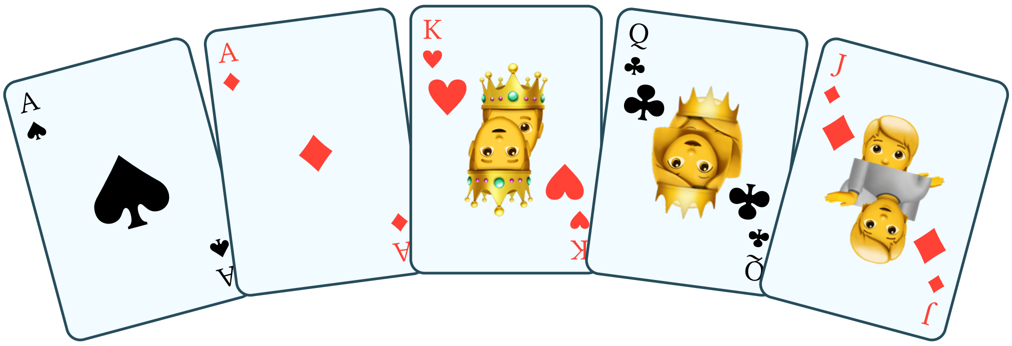
Conditional Probability: How Cards Change Odds

Game State

Player Hands

Your Hand:

After Flop:



Additional Content

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# Conditional Probability: How Cards Change Odds

## The Concept

Conditional probability asks: \"What's the probability of X, given that Y has already
happened?\"

In poker, every card dealt changes the probabilities for all remaining cards.

## Example: Drawing Aces

### Starting Situation
You're dealt two aces. Great hand! But what are your odds of improving?

### After the Flop
Three community cards are dealt. They don't help you. Now:
- **Cards remaining**: 47 (52 - 2 hole cards - 3 flop cards)
- **Aces remaining**: 2 (4 total - 2 in your hand)
- **Probability of getting another ace**: 2/47 = 4.26%

### The Key Insight
The probability changed! It's no longer 4/52 (7.69%) because:
1. You already have 2 aces
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2. 5 cards are already dealt
3. The deck has changed

Mathematical Formula

P(A | B) = P(A and B) / P(B)

Where:
- A = Event you're interested in (getting an ace)
- B = Condition that's already happened (you have 2 aces, flop is dealt)

Real-World Application

This is why card counting works in blackjack:
- As cards are dealt, the composition of the remaining deck changes
- High cards remaining = better odds for player
- Low cards remaining = better odds for house

The cards don't have memory, but the deck composition does change!